Adjustment and external replication of drug–hearing-loss signals in FAERS: a disproportionality analysis of FAERS and Canada Vigilance

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Abstract

Introduction: Recent disproportionality analyses of the FDA Adverse Event Reporting System (FAERS) reported many drug–hearing-loss signals absent from labels. We assessed which survive adjustment for hearing-specific biases and external replication.

Methods: Disproportionality analysis of 19.9 million deduplicated FAERS reports (2004–2026) and 1.2 million Canada Vigilance reports, following a protocol committed before analysis. For hearing loss and tinnitus, crude reporting odds ratios (ROR; signal: lower 95% bound > 1, ≥ 3 cases) were compared with Mantel–Haenszel estimates stratified by age, sex and period after excluding ear-related indications, reports naming established ototoxic drugs and lawyer reports. Methods were compared on a label-derived reference set (area under the ROC curve, AUC) and by replication in Canada Vigilance, observed and power-calibrated; robust unlabelled signals underwent case-by-case review.

Results: Adjustment changed the hearing-loss signal list (300 crude, 310 adjusted, 63 lost) but not agreement with labels (AUC 0.718 vs 0.723). Labelled signals replicated in Canada as often as predicted (observed/expected 1.00); unlabelled signals did not (0.42; post hoc, both databases adjusted: 0.20). Solicited patient-support-programme reports and shared co-medication produced spurious replications. Of 22 robust unlabelled hearing-loss and tinnitus signals (post hoc), 18 were attributed to report clusters, stimulated reporting, indication or an established class; 4 remained, led by pregabalin.

Conclusions: Most unlabelled FAERS hearing-loss signals do not survive adjustment, adjusted replication and provenance review. These findings are hypothesis-generating; cross-database replication needs adjusted estimates and explicit handling of solicited reports.

Key points

- Adjusting FAERS hearing-loss screens for indication, ototoxic co-medication and stimulated reporting changes which drugs are flagged but not their agreement with drug labels.
- In Canada Vigilance, labelled hearing-loss signals replicated as often as predicted, unlabelled

ones less than half as often; solicited patient-support-programme reports and shared co-medication produced spurious replications.

- After adjusted replication and case-by-case review of report provenance, 4 of 22 robust unlabelled signals remained, led by pregabalin; they are hypotheses for exposure-based studies.

1 Introduction

Drug-induced hearing loss ranges from reversible (salicylates, loop diuretics, quinine, macrolides) to permanent (aminoglycosides, platinum agents), and the classical ototoxic drug classes were identified mainly through clinical observation and case series [1]. For newer drugs, post-marketing spontaneous reports are often the first source of evidence, and recent studies have screened the FDA Adverse Event Reporting System (FAERS) across all drugs for hearing-loss signals [2–4] and for tinnitus [5, 6]; national databases in Japan and France have been used for the same purpose [7, 8].

The FAERS screens share a design: crude disproportionality across all reports, with novelty defined as absence from the label. In this setting crude disproportionality is open to biases specific to hearing outcomes. Corticosteroids, vasodilators and antivirals are prescribed *for* sudden hearing loss and tinnitus, and several diseases cause hearing loss themselves, so their treatments co-occur with the event (confounding by indication). A drug routinely given with a strongly ototoxic one—an antiemetic with cisplatin, an antibiotic with an aminoglycoside—inherits part of the association (co-medication or "innocent bystander" bias). Reporting can be stimulated by safety communications, litigation or media attention [9]. Editorials have questioned the value of disproportionality analyses that do not address such biases [10], and few FAERS screens test their signal lists against an independent data source or a reference set, as recommended for signal-detection methods [11].

We therefore asked a narrower question than "which drugs are ototoxic?": *which hearing-loss and tinnitus signals remain after adjustment for these biases, and do the signals that remain replicate in an independent national database and over time?* We compared crude and adjusted detection against a reference set derived from US labels, replicated every signal in Canada Vigilance, re-tested the signals reported by previous FAERS screens, and reviewed the report provenance of every robust unlabelled signal.

2 Methods

The study is reported according to the READUS-PV guideline [12, 13]; the checklist is given in Supplementary Table S1. The analysis protocol, with all definitions and thresholds of the primary analysis, was written and committed to a version-controlled repository (git) before the first result was computed; every later change is dated and labelled in that file (`PROTOCOL.md`), and analyses added after results were seen are reported as post hoc.

2.1 Data sources

We used all 90 quarterly ASCII extracts of the legacy Adverse Event Reporting System (2004Q1–2012Q3) and FAERS (2012Q4–2026Q2), published by the US Food and Drug Administration; the Canada Vigilance Adverse Reaction Online Database extract (Health Canada; reports received 1965 to 30 June 2026); and the openFDA bulk export of structured product labels (23 October 2026) [14]. Events are coded in MedDRA preferred terms (PTs), at the version current when each quarter was released (FAERS) and version 29.0 (Canada Vigilance). All data are public and de-identified.

2.2 Pre-processing and deduplication

In FAERS we kept the most recent version of each case (by FDA receipt date, preferring FAERS over legacy records, then case version), removed cases on FDA deletion lists, removed report identifiers recorded under two case numbers, and collapsed content duplicates: reports with identical sex, age, event date, country, set of suspect ingredients and set of reactions (applied only when event date and age were present). Drug names were mapped to active ingredients: the `prod_ai` field was split into ingredients; salt and hydrate words were removed when the remaining base was itself an observed ingredient; free-text names in records without `prod_ai` were mapped to the dominant ingredient recorded for that name ($\geq$70% of $\geq$5 records); INN/USAN synonyms were merged (for example glycopyrronium and glycopyrrolate); four-letter biologic suffixes were removed; and non-specific entries (for example "vitamins", "minerals", "... NOS") were dropped. Combination products contribute to each of their ingredients. In Canada Vigilance, reports flagged as duplicates of a report present in the extract were removed and product ingredients were normalised with the same rules.

2.3 Outcomes, exposure and comparator

The primary outcome, hearing loss, comprised the MedDRA PTs Deafness; Deafness unilateral, bilateral, neurosensory, transitory, permanent and mixed; Hypoacusis; Neurosensory hypoacusis; Sudden hearing loss; Ototoxicity; Hearing impaired; Hearing disability; and High frequency hearing loss. Conductive and congenital deafness were excluded. The secondary outcome was Tinnitus. An ingredient was exposed when reported as a suspect drug (FAERS primary suspect, secondary suspect or interacting; Canada suspect or interacting). The comparator for each ingredient was all other reports in the same population. Missing age and sex formed their own strata; missing indication or country was treated as not meeting the corresponding exclusion.

2.4 Disproportionality

For each ingredient we computed the reporting odds ratio (ROR) with a Woolf 95% confidence interval [15, 16] and the information component (IC) with its lower credibility bound IC_{025} [17]. A *crude signal* required at least three cases, a lower ROR bound above 1 and $IC_{025} > 0$.

The pre-specified *adjusted* analysis was run in a restricted population and stratified. We excluded (i) reports in which any drug had an ear or hearing indication (deafness, hearing, tinnitus, ototoxicity, Ménière's disease, otitis, vertigo, vestibular, labyrinthine and cochlear terms); (ii) reports that mentioned, in any role, a drug from an a-priori list of 29 drugs from the established ototoxic classes [1] (aminoglycosides, capreomycin and vancomycin; cisplatin, carboplatin and oxaliplatin; loop diuretics; quinine and related antimalarials; aspirin; macrolides; deferoxamine; vincristine; eflornithine) other than the drug under study; and (iii) FAERS reports submitted by lawyers (Canada Vigilance does not record this). For a drug on the list, reports naming the drug itself in a non-suspect role were also removed from its population. We then computed the Mantel–Haenszel (MH) OR [18] over 60 strata (five age bands $\times$ sex [male, female, unknown] $\times$ four reporting periods) with the Robins–Breslow–Greenland variance [19]. An *adjusted signal* required at least three cases in the restricted population and a lower 95% bound above 1. The implementation reproduces `statsmodels` estimates on simulated data to nine decimal places.

2.5 Reference set

From single-ingredient US labels we classified each ingredient as label-positive when a hearing-loss term appeared in a safety section (boxed warning, warnings, precautions, adverse reactions, post-marketing experience, contraindications, overdosage, drug abuse and dependence, or the corresponding over-the-counter sections) of at least half of its prescription labels, and label-negative when no label mentioned hearing loss or tinnitus; other ingredients were left out. Phrases that contain a hearing word without describing an adverse effect (for example the disease name "maternally inherited diabetes and deafness" and telephone lines for deaf users) were removed first. Discrimination was measured by the area under the ROC curve (AUC) among ingredients with $\geq$1,000 suspect FAERS reports, comparing the crude and adjusted lower 95% bounds (the like-for-like pair) and, secondarily, the IC_{025} and point estimates; the difference was bootstrapped (2,000 resamples). Because labels are partly informed by spontaneous reports, the reference set compares methods with each other; it does not estimate absolute accuracy.

2.6 Replication

For external replication, signals were derived in FAERS after removing reports of Canadian origin and tested in Canada Vigilance. Pre-specified rule: an ingredient was testable when the observed or expected number of Canadian cases was at least three, and replicated when the Canadian lower ROR bound exceeded 1 with at least three cases. The same rule applied to label-negative ingredients that were neither crude nor adjusted FAERS signals gave the background rate of replication without a known association; label-negative ingredients that *were* adjusted signals form a second reference group. Because the testability rule depends on the observed count, we also computed a power-calibrated comparison (post hoc): for every ingredient with a Canadian expected count $E > 0$, replication was defined as an observed count of at least a_{crit}, the smallest count ≥ 3 with Poisson $P(X \geq a \mid E) < 0.025$, and the expected number of replications was $\sum_i P(X \geq a_{\mathrm{crit},i} \mid E_i \times \mathrm{OR}_i)$,

with OR_i the adjusted FAERS estimate; observed-to-expected ratios near 1 indicate that signals replicate as often as their FAERS strength predicts. For temporal replication, signals derived from reports received up to 2014 were tested in reports received from 2015 onwards. Rates were compared with Fisher's exact test.

2.7 Post-hoc and sensitivity analyses

After the primary results were seen we added: Canada Vigilance without solicited reports ("Study" report type, which includes manufacturers' patient-support programmes); an extended indication filter (further ear terms; diseases that themselves cause hearing loss, such as Fabry disease, mucopolysaccharidoses, cryopyrin-associated syndromes, meningitis, giant cell arteritis and thyroid eye disease; and reports of prenatal exposure); and replication in which the Canadian estimate is adjusted in the same way as the FAERS one (*adjusted-to-adjusted*). In this post-hoc framework, ototoxic co-medication was handled in three further ways: as a stratum dimension instead of an exclusion (because restricting on a variable that reporters may add after observing hearing loss can introduce selection bias), with a short list of strongly ototoxic drugs only, and after excluding reports that list 15 or more drugs (report clusters). Canada Vigilance was also restricted to reports received from 2004.

2.8 Robust signals and case-level review

A *robust signal* (post hoc) was an adjusted FAERS signal that replicated adjusted-to-adjusted in Canada Vigilance without solicited reports. Robust signals whose event appears on no US label of any product containing the drug (single-ingredient or combination) were reviewed. For each we profiled the FAERS case series (without Canadian reports): share of cases from the most frequent country and receipt year, consumer reports, reports listing $\geq$15 drugs, reports co-naming an established ototoxic drug, prenatal exposure and leading indications; we searched PubMed for the drug with hearing terms. Each signal was assigned the first applicable category of a written rubric: K, established ototoxic class or association documented by a prospective or cohort clinical series; A, report cluster ($\geq$50% of cases in reports listing $\geq$15 drugs); B, ototoxic co-medication ($\geq$50% of cases co-name an established ototoxic drug); C, stimulated or solicited reporting documented for the drug; D, main indication itself associated with the event; E, none of these. The author applied the rubric, and an independent AI assessor applied it blind to the author's assignments, with the same profiles and literature access; agreement is reported as Cohen's κ.

2.9 Software and use of AI tools

Analyses used Python 3.10 (numpy 2.2.6, pandas 2.3.3, scipy 1.15.3, statsmodels 0.15.0, matplotlib 3.10.9). Data processing code, analyses and the first draft of this manuscript were produced with an AI assistant (Claude, Anthropic) working under the author's direction; three AI reviewers (methods, clinical, editorial) critiqued the draft, and their findings were addressed as described in the protocol

file. The author reviewed the code, results and text and is responsible for them.

3 Results

3.1 Data

The 90 FAERS extracts held 24,812,424 report versions, from which 20,692,682 cases remained after keeping the latest version; 104,186 cases on FDA deletion lists, cases without a reaction or mappable ingredient, 1,681 report identifiers recorded under two cases and 510,312 content duplicates were then removed, leaving **19,876,330 reports** (4% of drug rows could not be mapped; Supplementary Figure S1). They contained 83,883 hearing-loss and 42,900 tinnitus reports. Hearing-loss reports came more often from consumers and from patients aged ≥65 years than other reports (Supplementary Table S4). The primary adjusted population excluded 19,277 reports with an ear or hearing indication, 1,842,774 reports naming a listed ototoxic drug and 486,359 lawyer reports, leaving 17,571,385 reports (70,438 hearing-loss cases). Canada Vigilance contributed 1,210,857 reports after removing 58,623 duplicates (4,000 hearing-loss and 3,284 tinnitus reports).

Drugs of the established ototoxic classes gave strong signals in both databases (Supplementary Table S2), for example amikacin (FAERS adjusted OR 12.7 (11.6–13.9)), gentamicin (11.4 (9.8–13.2)) and cisplatin (4.3 (4.0–4.6)); so did the PDE5 inhibitors sildenafil (3.9 (3.7–4.2)) and tadalafil (4.2 (3.9–4.6)), whose labels carry hearing-loss warnings, and teprotumumab (59.8 (55.4–64.5)). Several label-positive drugs did not signal: oxaliplatin (crude ROR 0.44), vardenafil (6 cases), and, after adjustment, carboplatin (crude 1.96, adjusted 0.82) and vincristine, whose crude associations were carried largely by reports that also named another listed drug—the sensitivity that the comedication filter costs.

3.2 Adjustment changes the signal list but not its agreement with labels

In FAERS without Canadian reports the crude rule flagged 300 ingredients for hearing loss and the adjusted rule 310; 63 crude signals were lost and 73 appeared (Table 1). Against the reference set (106 label-positive and 927 label-negative ingredients with ≥1,000 suspect reports), discrimination was moderate and practically identical for the crude and adjusted lower bounds (AUC 0.718 and 0.723; difference 0.005, 95% CI -0.023 to 0.030, which excludes an improvement larger than 0.03). The same held for point estimates (0.700 vs 0.705), for the IC_{025} (0.720), at lower report thresholds, without the 29 listed drugs, and for tinnitus (0.775 vs 0.777). Adjustment lowered sensitivity (0.40 to 0.35) with little change in specificity (0.866 to 0.873).

3.3 Replication in Canada Vigilance

Under the pre-specified rule, adjusted hearing-loss signals already on a label replicated in 34/50 (68%) of testable cases, unlabelled adjusted signals in 31/76 (41%), label-negative adjusted signals in 21/54 (39%), crude-only signals in 9/21 (43%), and the background of label-negative non-signals

in 43/214 (20%) (Table 1). The power-calibrated comparison separated the groups more clearly (Figure 1): labelled signals replicated exactly as often as their FAERS strength predicted (31 observed vs 31.2 expected (O/E 1.00)), whereas unlabelled signals replicated less than half as often (25 observed vs 59.5 expected (O/E 0.42)), like label-negative signals (18 observed vs 39.0 expected (O/E 0.46)). Tinnitus showed the same contrast (60 observed vs 64.4 expected (O/E 0.93) for labelled and 22 observed vs 49.4 expected (O/E 0.45) for unlabelled signals).

The background replicated more often than chance (36 observed vs 8.5 expected without any association). Solicited reports explained much of this: *Study* reports, which include manufacturers' patient-support programmes, made up 61% of Canadian hearing-loss cases for the background drugs that replicated, against 14% for replicated labelled signals and 36% overall; for ocrelizumab, mepolizumab and omalizumab the share was 93%, 93% and 86%, against 1% for gentamicin and 1% for cisplatin. The mirror image appeared in FAERS: tafamidis (adjusted OR 4.3 (3.8–4.8)) had 361 hearing-loss reports in FAERS but none in Canada Vigilance against 3.4 expected, and sacubitril (2,801 FAERS reports) none against 1.9 expected (too few to be testable under the pre-specified rule).

Co-medication bias also replicated when crude estimates were compared. Etoposide, given in cisplatin-containing regimens, was a crude signal in both databases (FAERS ROR 2.09 (1.91–2.28); Canada 5.13 (3.68–7.14)) and disappeared after adjustment in both (FAERS 0.67 (0.46–0.98); Canada 1.15 (0.29–4.64)). Temporal replication within FAERS did not separate crude-only from adjusted hearing-loss signals (early adjusted signals recurred after 2014 in 81/155 (52%), crude-only signals in 18/28 (64%), background 137/938 (15%)); for tinnitus adjusted signals recurred more often (100/121 (83%) vs 9/15 (60%)).

3.4 Post-hoc analyses

With the extended indication filter, Canada Vigilance restricted to unsolicited reports and both estimates adjusted (Table 2), labelled hearing-loss signals replicated in 26/44 (59%), unlabelled signals in 10/51 (20%) and the background in 19/158 (12%) (unlabelled vs background, Fisher $p = 0.24$); calibrated O/E ratios were 0.81 for labelled and 0.20 for unlabelled signals. Handling co-medication by stratification instead of exclusion, with a short list of strongly ototoxic drugs, or after removing reports listing $\geq$15 drugs left this contrast unchanged (unlabelled O/E 0.20–0.27 across variants, labelled 0.81–1.02), as did restricting Canada Vigilance to reports from 2004.

3.5 Signals from previous screens

Of the 23 hearing-loss signals that Li et al. classified as unlabelled [2], 16 were crude and 16 adjusted signals in FAERS without Canadian reports, 6 were testable in Canada and 2 replicated; under the post-hoc definition 1 remained (glycopyrrolate). Their strongest unlabelled signals included sacubitril/valsartan and tafamidis. Of the 28 unlabelled paediatric signals of Liu et al. [4], 11 were adjusted signals in our all-age data and 1 met the post-hoc definition (azithromycin, which is on the

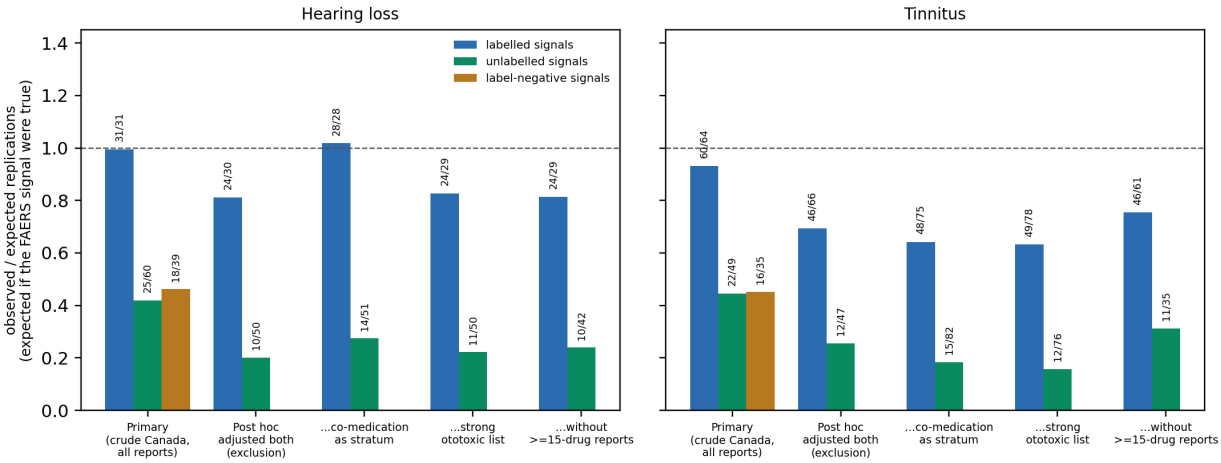


Figure 1: Power-calibrated replication in Canada Vigilance: observed replications divided by those expected if each FAERS adjusted estimate were true (labels give observed/expected counts). Primary: pre-specified signals, Canada crude estimates, all reports. Post hoc: both databases adjusted, Canada without solicited reports, four ways of handling co-medication. A ratio near 1 means signals replicate as their FAERS strength predicts.

a-priori ototoxic list); cefixime, ceftriaxone, etoposide, isoniazid were lost on adjustment (Table 3). Only abstract-level lists were available for the remaining screens (Supplementary Table S6).

3.6 Case-level review

Under the post-hoc definition, 10 hearing-loss and 12 tinnitus signals were robust and absent from every US label (Table 4, Figure 2). The review attributed 4 to report clusters (for example candesartan–tinnitus: 80% of cases from the United Kingdom, median 42 drugs per report), 6 to stimulated or mass-exposure reporting (mRNA COVID-19 vaccines, rofecoxib, finasteride), 7 to an indication itself associated with the event (rheumatoid arthritis for etanercept and tofacitinib; COPD for glycopyrrolate; glaucoma for travoprost; diabetes for insulin aspart; migraine for galcanezumab; hypothyroidism for liothyronine) and 1 to an established ototoxic class (oral paromomycin). Four remained without a dominant artefactual explanation: pregabalin and amlodipine for hearing loss, and doxycycline and chlorthalidone (10 cases) for tinnitus. The independent assessor agreed on 19 of 22 assignments (Cohen's $\kappa = 0.82$); it placed Tozinameran (hearing loss) in K rather than C; Amlodipine (hearing loss) in D rather than E; Galcanezumab (tinnitus) in E rather than D (Supplementary Table S3), which would leave the number of unexplained signals at 4.

4 Discussion

Three findings matter for anyone screening spontaneous reports for ototoxicity. First, adjusting for indication, ototoxic co-medication, lawyer reports and demographics changed a substantial part of the hearing-loss signal list (63 of 300 crude signals lost, 73 gained) but did not improve agreement

Table 1: Crude versus adjusted detection (pre-specified analyses). Replication counts are replicated/testable ingredients.

	Crude	Adjusted
Hearing loss		
Reference set, positive/negative ingredients	106/927	
AUC, lower 95% bound	0.718	0.723
difference (95% CI)	0.005 (-0.023 to 0.030)	
Sensitivity / specificity of the signal rule	0.40 / 0.87	0.35 / 0.87
Signals (FAERS without Canadian reports)	300	310
Replicated in Canada[a]: labelled adjusted signals	34/50 (68%)	
unlabelled adjusted signals	31/76 (41%)	
label-negative adjusted signals	21/54 (39%)	
crude-only signals	9/21 (43%)	
background (label-negative non-signals)	43/214 (20%)	
Observed/expected replications[b]: labelled	31/31.2 = 1.00	
unlabelled	25/59.5 = 0.42	
label-negative	18/39.0 = 0.46	
Tinnitus		
Reference set, positive/negative ingredients	186/927	
AUC, lower 95% bound	0.775	0.777
difference (95% CI)	0.002 (-0.015 to 0.019)	
Sensitivity / specificity of the signal rule	0.51 / 0.85	0.50 / 0.87
Signals (FAERS without Canadian reports)	399	349
Replicated in Canada[a]: labelled adjusted signals	65/89 (73%)	
unlabelled adjusted signals	24/56 (43%)	
label-negative adjusted signals	17/43 (40%)	
crude-only signals	13/30 (43%)	
background (label-negative non-signals)	31/193 (16%)	
Observed/expected replications[b]: labelled	60/64.4 = 0.93	
unlabelled	22/49.4 = 0.45	
label-negative	16/35.5 = 0.45	

[a] Pre-specified rule: FAERS without Canadian reports; Canada crude ROR, all reports. [b] Power-calibrated (post hoc): observed replications vs those expected if each FAERS adjusted estimate were true.

with labels. Second, replication in an independent national database behaved as a validation test should for labelled associations—they replicated as often as their strength predicted—but unlabelled signals replicated less than half as often as predicted, about as often as signals for drugs whose labels are silent on hearing. Third, replication is not a clean validator on its own: solicited patient-support-programme reports produced "replications" for drugs with no known association, and co-medication bias replicated whenever crude estimates were compared. After adjusted replication and a review of report provenance, 4 of 22 robust unlabelled signals remained.

These results qualify the previous all-drug screens [2–4] rather than contradict them. Their labelled signals largely held; their unlabelled signals mostly did not, and several of the strongest are consistent with how reports were collected rather than with pharmacology. Tafamidis and sacubitril/valsartan are examples: thousands of consumer reports in FAERS, consistent with manufacturer programmes in the United States, and none in Canada. The same mechanism produced spurious

Table 2: Post-hoc analyses: extended indication filter, Canada Vigilance without solicited reports, both estimates adjusted, with four ways of handling ototoxic co-medication. O/E: observed/expected replications. Robust: adjusted signals replicated adjusted-to-adjusted (unlabelled in brackets).

Co-medication handling	Signals	AUC	Labelled repl.	Unlabelled repl.	Background repl.	O/E lab./unlab.	Robust
Hearing loss							
Exclusion (as primary)	295	0.719	26/44 (59%)	10/51 (20%)	19/158 (12%)	0.81 / 0.20	36 (10)
Co-medication as stratum	323	0.718	28/43 (65%)	17/57 (30%)	28/158 (18%)	1.02 / 0.27	45 (17)
Strong ototoxic list only	303	0.714	26/43 (60%)	14/54 (26%)	22/158 (14%)	0.83 / 0.22	40 (14)
Without reports listing ≥15 drugs	269	0.725	26/41 (63%)	8/48 (17%)	5/159 (3%)	0.81 / 0.24	34 (8)
Tinnitus							
Exclusion (as primary)	346	0.776	52/89 (58%)	12/50 (24%)	8/153 (5%)	0.69 / 0.26	64 (12)
Co-medication as stratum	434	0.762	59/96 (61%)	20/66 (30%)	15/154 (10%)	0.64 / 0.18	79 (20)
Strong ototoxic list only	410	0.769	54/97 (56%)	12/60 (20%)	14/154 (9%)	0.63 / 0.16	66 (12)
Without reports listing ≥15 drugs	317	0.786	47/86 (55%)	7/45 (16%)	1/153 (1%)	0.75 / 0.31	54 (7)

Table 3: Previously reported FAERS hearing-loss signals re-tested here (FAERS without Canadian reports). Columns: reported, crude signal, adjusted signal, adjusted and testable in Canada, adjusted and replicated, robust (post hoc).

Study	Authors' label status	Reported	Crude	Adjusted	Testable	Replicated	Robust
Li et al. 2024 (all ages)	labelled	18	18	15	10	10	8
Li et al. 2024 (all ages)	unlabelled	23	16	16	6	2	1
Liu et al. 2025 (children)	labelled	29	16	15	13	11	11
Liu et al. 2025 (children)	unlabelled	28	14	11	9	7	1

Canadian "replications" for biologics. Solicited reports are known to distort disproportionality [20], and their share differs between databases, so report provenance can manufacture both a signal and its replication. Bias that is shared between databases—co-prescription in standard regimens—also replicates; etoposide is the clearest example here.

Two inexpensive steps addressed most of what we found: comparing *adjusted* estimates in both databases, and profiling each case series for concentration—solicited or consumer reports, a single country or year, very drug-dense reports. The power-calibrated comparison is also simple and avoids the outcome-dependent testability of rules based on observed counts; it could be extended with empirical calibration against negative controls [21] or with regression-based adjustment for co-reported drugs [22]. We suggest that screens for hearing outcomes report these provenance indicators for every signal, in line with the case-by-case item of READUS-PV [12, 13].

The remaining candidates are hypotheses with plausible alternative explanations. Pregabalin, an $\alpha_2\delta$ ligand, had 1,703 FAERS hearing-loss reports (adjusted OR 3.25 (3.09–3.43)) that replicated in Canada; a case of reversible, dose-related hearing loss has been described [23], but diabetes, neuropathy and age confound [24], and its labels list tinnitus and hyperacusis. For amlodipine, whose labels mention tinnitus, a large cohort found a *lower* risk of sensorineural hearing loss with calcium-channel blockers [25], which argues against a causal effect. Doxycycline–tinnitus may reflect pulsatile tinnitus from drug-induced intracranial hypertension, which is labelled [26], or the infections it treats. None of these should change practice without studies that have exposure data.

Table 4: Robust signals absent from all US labels, with case-level review category: A report cluster; C stimulated or solicited reporting; D indication associated with the event; K established ototoxic class; E none of these (bold). FAERS cases: reports without Canadian origin naming the drug as suspect with the event. Consumer and ≥15 drugs: share of those cases. Hypotheses only, not evidence of causation.

Drug	Event	FAERS cases	FAERS adj. OR (95% CI)	Canada adj. OR (95% CI)	Consumer	≥15 drugs	Cat.
Rofecoxib	HL	243	2.39 (2.02–2.82)	4.42 (2.35–8.31)	7%	14%	C
Tozinameran	HL	57	2.36 (1.76–3.15)	7.04 (5.70–8.69)	72%	26%	C
Elasomeran	HL	20	2.36 (1.44–3.86)	3.30 (2.01–5.44)	75%	35%	C
Etanercept	HL	2095	1.15 (1.10–1.21)	2.28 (1.71–3.05)	49%	3%	D
Tofacitinib	HL	1728	3.27 (3.11–3.45)	1.39 (1.05–1.84)	82%	4%	D
Glycopyrrolate	HL	315	3.08 (2.74–3.45)	4.40 (1.82–10.67)	50%	3%	D
Insulin Aspart	HL	250	2.27 (2.00–2.58)	2.69 (1.11–6.50)	82%	5%	D
Travoprost	HL	140	3.71 (3.13–4.40)	4.96 (1.58–15.54)	75%	0%	D
Pregabalin	HL	1703	3.25 (3.09–3.43)	2.06 (1.24–3.44)	62%	12%	E
Amlodipine	HL	792	2.39 (2.21–2.57)	2.70 (1.65–4.44)	70%	9%	E
Candesartan	Tin	422	11.72 (10.40–13.22)	2.80 (1.68–4.68)	38%	83%	A
Propranolol	Tin	189	1.79 (1.40–2.28)	2.31 (1.03–5.17)	24%	66%	A
Tozinameran	Tin	109	6.53 (5.01–8.50)	10.11 (8.46–12.07)	46%	51%	A
Indapamide	Tin	44	2.05 (1.29–3.26)	3.44 (1.53–7.71)	18%	55%	A
Finasteride	Tin	293	9.73 (8.63–10.95)	3.04 (1.26–7.37)	60%	0%	C
Rofecoxib	Tin	189	2.53 (2.13–3.02)	2.26 (1.12–4.54)	8%	21%	C
Elasomeran	Tin	13	3.62 (1.94–6.74)	3.50 (2.26–5.43)	69%	31%	C
Galcanezumab	Tin	96	1.86 (1.52–2.28)	16.70 (6.73–41.48)	88%	0%	D
Liothyronine	Tin	40	4.90 (3.56–6.75)	16.73 (5.22–53.59)	78%	32%	D
Doxycycline	Tin	220	4.54 (3.88–5.31)	2.36 (1.12–4.97)	48%	6%	E
Chlorthalidone	Tin	10	1.94 (1.04–3.62)	4.57 (1.70–12.29)	90%	0%	E
Paromomycin	Tin	3	27.75 (8.63–89.29)	262.76 (53.09–1300.46)	67%	0%	K

Limitations. Spontaneous reports lack exposure denominators, and disproportionality measures reporting, not risk or incidence. The label reference set is imperfect: labels are partly informed by spontaneous reports, list rare events without causal assessment, and route-specific labels (for example topical gentamicin) blur ingredient-level status; we used it only to compare methods. Excluding reports with an ototoxic co-medication conditions on a variable that reporters may add after observing the event; stratifying on it instead gave the same conclusions. Indications are often missing, unreported co-treatment (for example aminoglycosides in mycobacterial disease) remains, and content deduplication cannot catch near-duplicate series, which the report-cluster indicator only partly addresses. The Canada Vigilance documentation does not state that the extract is domestic; restricting it to reports sent directly to Health Canada gave similar results (39/72 (54%) of adjusted signals replicated, background 12/58 (21%)). About 8,400 ingredients were screened without multiplicity correction; the replication and calibration analyses, rather than individual p-values, carry the conclusions. Several analyses—the solicited-report exclusion, the extended indication filter, adjusted-to-adjusted and calibrated replication, the co-medication variants and the case-level review—were added after the primary results were seen, partly in response to internal review; they are labelled post hoc and the pre-specified results are reported unchanged. The case-level review used a written rubric applied by the author and by an AI assessor, not by independent clinicians.

Conclusions. In FAERS, most drug–hearing-loss signals that are not already on a label do not survive adjustment, adjusted replication in Canada Vigilance and a review of report provenance; dis-

proportionality analysis remains a hypothesis-generating tool. Replication in a second spontaneous-reporting database is informative only when both estimates are adjusted and solicited reports are handled explicitly. The few remaining candidates, led by pregabalin, merit study with exposure-based designs.

Statements and declarations

Funding: no funding was received for this study. **Competing interests: Ethics approval:** not applicable; the study used public, de-identified data. **Consent to participate and for publication:** not applicable. **Data availability:** FAERS and Canada Vigilance data are publicly available from the FDA and Health Canada; result tables are in the repository below. **Code availability:** code, protocol (`PROTOCOL.md`, with commit history) and results: **Author contributions:** OT conceived and designed the study, directed and checked the analysis, and wrote and approved the manuscript.

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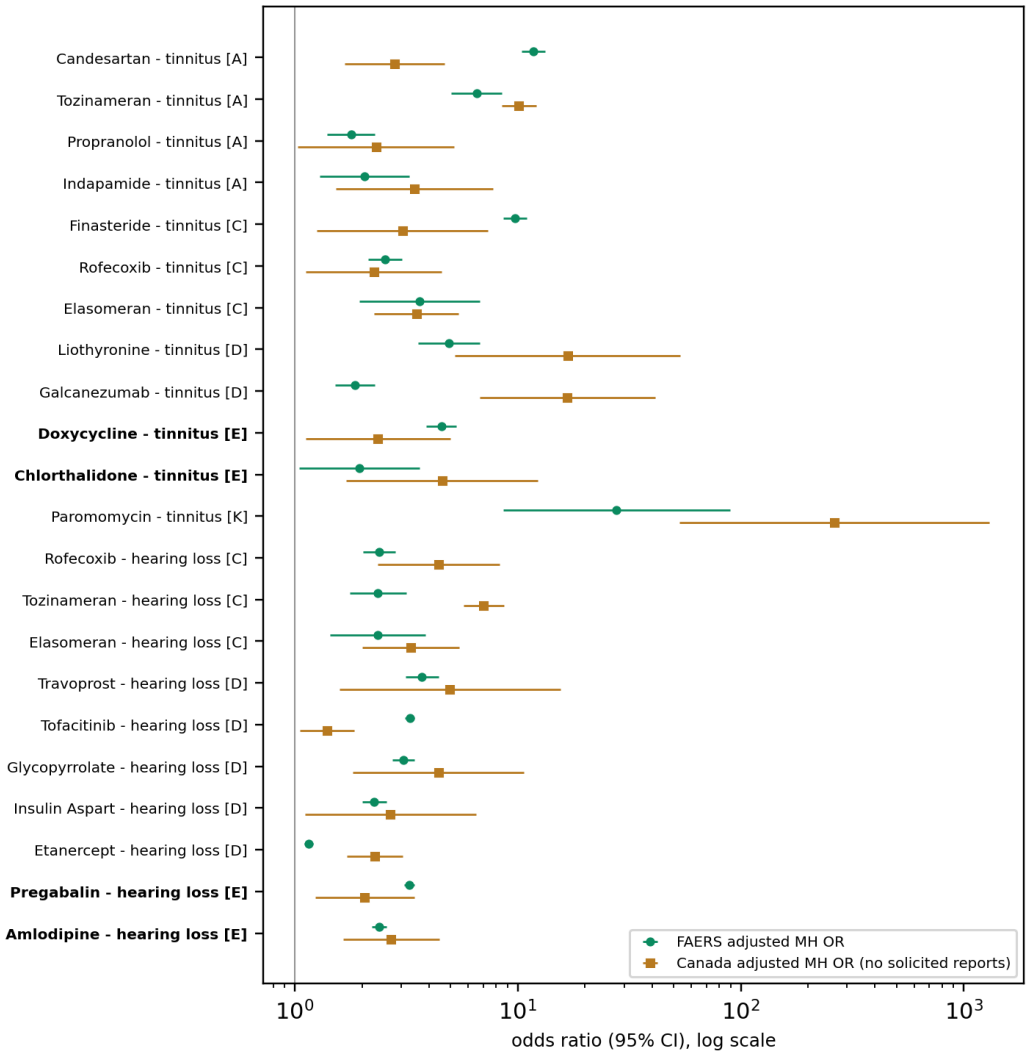


Figure 2: Robust signals absent from all US labels: FAERS and Canada Vigilance adjusted estimates, with case-review category (A report cluster; C stimulated reporting; D indication; K established class; E none, bold).